recsim-arena: How Recommender Mechanism Design Shapes Creator Equilibria

A Multi-Agent Reinforcement Learning Study of Bait, Quality, and Exposure Inequality

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github.com/ozeidan9/recsim-arena

June 2026

Abstract

Recommender systems are usually studied as fixed maps from users and a static item catalogue to slates. On real platforms the catalogue is not static: it is produced by strategic creators who adapt what they make to whatever the ranker rewards. I treat the recommender as a *mechanism* and the creators as *independent reinforcement-learning agents*, then ask how the choice of mechanism shapes the equilibrium that emerges. *recsim-arena* is a small, fully controllable market—50 users across 5 preference clusters, 20 creators learning with independent PPO (IPPO), 16-dimensional content, 200 interaction rounds per episode, and a slate of 5—in which a logit click model rewards relevance and persistent quality while penalising engagement-bait through accumulating user fatigue. I compare three mechanisms: a single-stage relevance-plus-bait ranker (M1), a two-stage retrieval-then-rerank pipeline (M2), and a quality-gated variant (M3) that inserts a quality gate before reranking. The headline result is robust across five seeds: the quality gate moves the creator equilibrium decisively toward quality (final bait roughly 0.03–0.09 under M3 versus 0.46–0.67 under M1; final quality roughly 0.87–0.97 versus 0.40–0.56). Under pure engagement ranking I find two effects of very different robustness—a robust collapse in exposure equality (Gini up ∼263% on average across all five seeds, s.d. 59) but only a seed-fragile rise in bait—and I report them as such rather than overclaiming a general "clickbait collapse." Two single-seed exploratory studies round out the picture: a two-stage pipeline concentrates exposure relative to single-stage (Gini 0.187 vs 0.092), and a recommender-temperature sweep shows that sharp ranking suppresses bait while flat ranking lets the bait term dominate. The quality gate here is an analytical linear stand-in for an LLM judge, not a live LLM; the scope is deliberately small. I argue the contribution is methodological as much as empirical: a reproducible testbed and an honest accounting of which mechanism-design effects survive re-seeding and which do not.

1 Introduction

Most academic treatments of recommendation hold the item catalogue fixed. You are given users, given items, and your job is to rank. That framing is convenient and, for a single serving decision, often correct. It is also increasingly at odds with how large platforms actually work. On YouTube, TikTok, Spotify, or a news feed, the catalogue is endogenous: creators decide what to make, and they decide based on what they believe the algorithm will surface and reward. The ranker is therefore not just choosing among content—over the medium term it is *shaping* the content that exists. If the ranker rewards watch-time bait, creators learn to make bait. If it rewards a narrow popular niche, creators crowd into that niche and the long tail starves.

This is a mechanism-design problem dressed up as a ranking problem, and the supply side has only recently received the attention it deserves. A line of game-theoretic work characterises the equilibria that strategic creators converge to under a given ranking rule (Ben-Porat & Tennenholtz, 2018; Jagadeesan et al., 2023; Hron et al., 2022; Immorlica et al., 2024). What is comparatively rare is a *learning* account: creators are rarely best-responders with closed-form utilities; they are adaptive agents groping toward whatever pays, and the platform's mechanism is the environment they adapt to. That is precisely the regime where multi-agent reinforcement learning is the natural tool, and where a simulator earns its keep—you cannot A/B test a counterfactual mechanism on a live creator economy without consequences, but you can in silico.

recsim-arena is my attempt at a clean, small, honest version of that experiment. Twenty creators are independent PPO learners competing for the attention of fifty users; each creator chooses a 16-dimensional content vector and a scalar "bait" level; the platform ranks, and users click according to a logit model in which relevance and quality help while bait helps in the short run but accrues fatigue. On top of this I vary the *mechanism*—single-stage, two-stage, and quality-gated—and watch where the creator population settles.

The paper makes three claims and is careful about the evidential weight behind each. First, a quality gate inserted into the ranking pipeline reliably shifts the creator equilibrium from bait toward quality; this is my strongest and most reproducible finding, confirmed across all five random seeds, and it anchors the paper. Second, pure engagement ranking produces two effects that are often conflated but behave very differently under re-seeding: a *robust* collapse in exposure equality, and a *seed-fragile* rise in bait. Reporting these together, with their different robustness, is itself a contribution—it is exactly the kind of distinction that gets lost when a simulation result is sold as a clean morality tale. Third, two smaller single-seed studies suggest that pipeline depth concentrates exposure and that recommender temperature governs how hard creators are pushed toward relevance-matching versus bait.

I want to be explicit about scope up front, because honesty about limitations is part of the point. The quality gate in my experiments is an analytical linear proxy for an LLM judge, not a live language model; the system is small (20 creators, 50 users); and two of the four studies use a single seed and therefore carry no error bars. I flag these throughout rather than burying them.

2 Related Work

Simulation platforms for recommendation. The closest infrastructure ancestors are RecSim (Ie et al., 2019) and its successor RecSim NG (Mladenov et al., 2021), which provide configurable—and in the NG case differentiable—environments for sequential recommendation and for modelling multiple agents (users, content providers, vendors) in an ecosystem. T-RECS (Lucherini et al., 2021) similarly offers an open-source Python simulator for the societal-impact questions that arise when an algorithm mediates between users and creators, replicating earlier simulation studies such as the algorithmic-confounding work of Chaney et al. *recsim-arena* is smaller and more opinionated than these: rather than a general framework, it is a focused arena for one question—how the *mechanism* reshapes a learning creator population—and it implements creators as full RL agents rather than as samplers from a fixed behaviour model.

Strategic behaviour, performativity, and creator incentives. The conceptual backbone is the literature on prediction that changes its own target. Strategic classification (Hardt et al., 2016) and the more general performative-prediction framework (Perdomo et al., 2020) formalise how agents move their features in response to a deployed model. In recommendation specifically, Ben-Porat & Tennenholtz (2018) initiated a game-theoretic treatment of strategic content providers. Jagadeesan et al. (2023) study supply-side equilibria when producers choose high-dimensional content vectors, showing that multi-dimensionality and heterogeneity create the potential for specialisation and genre formation at equilibrium. Hron et al. (2022) model creator incentives on algorithm-curated platforms. Yao and collaborators bound the welfare loss of top-k recommendation under competing creators (Yao et al., 2023a) and ask whether monotone reward mechanisms are always beneficial (Yao et al., 2023b, 2024). The most direct antecedent for my bait/quality split is Immorlica et al. (2024): they model a game in which engagement-based optimisation rewards both genuine quality and gaming tricks such as clickbait, finding that quality and gaming are positively correlated at equilibrium and that average consumed quality can *fall* as gaming becomes costlier. Boutilier et al. (2023) survey the broader programme of treating recommender ecosystems through the joint lens of mechanism design, reinforcement learning, and generative models. *recsim-arena* sits inside this programme but trades theoretical generality for a controlled, learning-based, mechanism-comparison experiment.

Multi-agent reinforcement learning. Creators here are trained with independent PPO. The relevant methodological references are PPO itself (Schulman et al., 2017); the observation by de Witt et al. (2020) that independent PPO (IPPO) can match or exceed state-of-the-art joint-learning approaches on SMAC with little tuning, attributed in part to IPPO's robustness to certain forms of environment non-stationarity; and the broader MAPPO study of Yu et al. (2022). My setting is competitive rather than cooperative and non-stationary by construction—each creator's environment includes 19 other learning creators plus, under M2/M3, an online-updating reranker—so I use IPPO as a pragmatic, well-understood learner rather than claiming convergence guarantees.

Two-stage architectures and feedback loops. Industrial recommenders are overwhelmingly two-stage: a fast nomination/retrieval step preselects candidates, then a slower, more accurate ranker refines them (Covington et al., 2016; Hron et al., 2021). Hron et al. (2021) show that the two stages interact in ways not explained by either component alone, deriving a generalisation lower bound under which independently trained nominators can perform on par with uniformly random recommendation. That directly motivates my interest in whether pipeline depth changes equilibrium exposure. Separately, feedback-loop analyses such as Jiang et al. (2019) on degenerate feedback loops, and the sales-diversity simulations of Fleder & Hosanagar (2009), show how recommend-then-learn cycles can drive concentration and homogenisation. My M2 study is a creator-side analogue: the reranker is trained online on the engagement it helps generate.

Exposure fairness and concentration metrics. I measure exposure inequality with the Gini coefficient over creator impressions, following a long tradition of using Gini and aggregate-diversity measures to quantify popularity bias and long-tail starvation. The normative target—exposure proportional to relevance or merit rather than winner-take-all—is the fairness-of-exposure framing of Singh & Joachims (2018) and the equity-of-attention framing of Biega et al. (2018). I treat a rising Gini as a warning sign of mechanism-induced concentration, not as a fairness verdict in itself.

LLM-as-judge and quality gating. My M3 mechanism inserts a quality gate before reranking. Conceptually this is a quality-gating step of the kind an LLM judge could implement, and the LLM-as-judge literature (Zheng et al., 2023; Gu et al., 2024) documents both the promise and the well-known biases—position, verbosity, self-enhancement—of using language models as evaluators. I deliberately do *not* run a live LLM here; M3 uses an analytical linear gate as a stand-in (§3). The norms of this kind of work—treating short-term engagement as a flawed proxy and taking the long-term, multi-stakeholder consequences of sequential decisions seriously—are those articulated by the CONSEQUENCES workshop series at RecSys (Jeunen et al., 2022), which I take as the relevant research community for this paper.

3 Methods

3.1 Environment

The market has 50 users partitioned into 5 preference clusters and 20 creators. Each user and each piece of content is a vector in a 16-dimensional latent space (`content_dim` = 16). An episode runs 200 interaction rounds; at each round the mechanism assembles a slate of `slate_size` = 5 items per user from the creators' current content, the user issues clicks, and creators collect reward. Users have a latent preference vector; creators each control a content vector c and a scalar bait level.

3.2 Click model

A user with preference vector u evaluates a candidate with content c, quality q, and bait level b through the logit

$$\text{logit} \;=\; \frac{u \cdot c}{\tau} \;+\; \alpha_{\text{quality}}\, q \;-\; \beta_{\text{bait}}\, b \cdot \text{fatigue}, \tag{1}$$

where τ is a recommender temperature scaling the relevance term. Quality contributes a persistent positive term. Bait contributes positively to raw attractiveness but is multiplied by an accumulating fatigue term, so leaning on bait pays in the short run and is taxed as users tire of it. This is a deliberately simple operationalisation of the clickbait-versus-quality tension formalised by Immorlica et al. (2024): quality is effort that helps users; bait is effort that extracts a click now at a cost to satisfaction later.

3.3 Mechanisms

I compare three platform mechanisms holding the environment fixed.

M1 — `SingleStageMechanism` (baseline). A single deterministic top-k ranker scoring items by relevance plus a `bait_weight` term. This is the "pure engagement ranker" against which the others are compared.

M2 — `TwoStageMechanism`. A two-tower retrieval stage (implemented with NumPy inner-product search) nominates candidates, which are then scored by an online-trained engagement reranker (an MLP updated during the episode on the engagement it observes). This mirrors the retrieval-then-ranking structure of industrial systems and lets me ask whether pipeline depth, per se, changes equilibrium exposure.

M3 — `LLMGateMechanism`. M2 plus a quality gate inserted *before* reranking. In the experiments reported here the gate is a `SimpleLinearGate` with $\alpha_{\text{quality}} = 3$ and $\alpha_{\text{bait}} = -3$—an analytical linear proxy that rewards quality and penalises bait, standing in for what a learned LLM judge might score. The codebase also contains a live `LLMQualityGate` and a `DistilledGate`, but **neither was executed for this paper**; all M3 results are from the linear gate. I therefore describe M3 as a quality gate / linear LLM-judge stand-in and never as live-LLM results.

Table 1: Summary of findings and their evidential weight. Multi-seed studies use seeds $\{42, 0, 7, 13, 99\}$.

Study	Finding	Headline numbers	Robustness
H2 (M3 vs M1)	Quality gate ⇒ quality equilibrium	bait 0.03–0.09 vs 0.46–0.67; quality 0.87–0.97 vs 0.40–0.56	5/5 seeds (confirmed)
H1a (M1)	Exposure (Gini) collapse	mean +263% (s.d. 59); all > +50%	5/5 seeds
H1b (M1)	Bait inflation	+44% on seed 42 only; decreases on seeds 7, 13, 99	1/5 seeds (fragile)
H4 (M2 vs M1)	Pipeline depth concentrates exposure	Gini 0.187 vs 0.092 (∼ +203%)	seed 42 only
H3 (M1, τ sweep)	Temperature governs bait	bait $0.12 \to 0.82 \to\approx 0.99$ ($\tau = 0.1 \to 1 \to 3, 10$)	seed 42 only

3.4 Creator learning (IPPO)

Each creator is an independent PPO learner (Schulman et al., 2017; de Witt et al., 2020) optimising its own click-based reward by adjusting its content vector and bait level. There is no parameter sharing and no centralised critic; from any one creator's perspective the environment—19 other learning creators and, under M2/M3, an online-updating reranker—is non-stationary. I use IPPO precisely because it is a strong, simple, and well-characterised baseline in that regime rather than because I can certify equilibrium convergence.

3.5 Metrics

Per-episode logs (`ippo_log.json`) record `diversity_entropy` (entropy of the content/topic distribution), `mean_quality`, `mean_bait`, `gini` (Gini coefficient of creator exposure), `ild` (intra-list diversity), reward terms (`mean_ep_reward`, `total_ep_reward`), and PPO diagnostics (`policy_loss`, `value_loss`, `entropy`), indexed by `episode`. The two outcomes I lean on hardest are `mean_bait` (the population's reliance on engagement-bait) and `gini` (exposure inequality across creators).

4 Experimental Setup

Each experiment fixes the environment (50 users / 5 clusters / 20 creators / `content_dim`= 16 / 200 rounds / `slate_size`= 5) and varies one factor—the mechanism (H1, H2, H4) or the recommender temperature (H3). Every run directory carries a `config.json` recording the environment, IPPO, and mechanism parameters together with the git commit and seed, for reproducibility. The multi-seed studies (H1, H2) use the five seeds $\{42, 0, 7, 13, 99\}$; the exploratory studies (H4, H3) use seed 42 only. Retrieval in M2/M3 uses NumPy inner-product search rather than a FAISS index; this was a deliberate engineering choice to avoid a PyTorch+FAISS OpenMP crash on macOS, and at this scale (20 creators) exhaustive inner-product search is mathematically equivalent to an approximate-nearest-neighbour index. I report metrics directly from the saved JSON logs and label single-seed results as exploratory.

5 Results

I order the results by evidential weight, leading with the most robust finding. Table 1 summarises.[1]

[1] Figures are the project's own `paper/figures/` outputs; shaded bands in the multi-seed panels (H1, H2) show across-seed spread. Headline numbers in the text were verified against the committed run summary; for full reproducibility the per-episode `ippo_log.json` traces can be re-read from the repository.

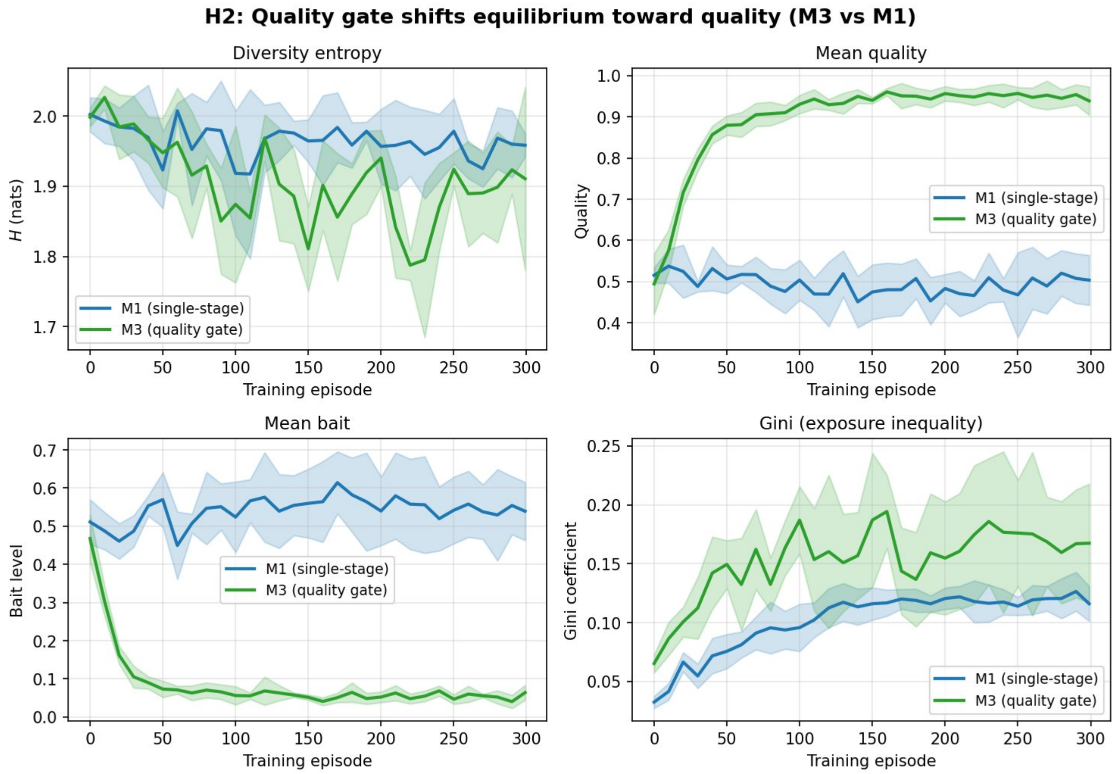

Figure 1: H2 (headline, 5 seeds, shaded = across-seed spread). Quality gate (M3, green) versus single-stage engagement ranking (M1, blue). Bottom-left: M3 drives mean bait down to ∼ 0.03–0.09 while M1 holds at ∼ 0.46–0.67. Top-right: M3 lifts mean quality to ∼ 0.87–0.97 versus ∼ 0.40–0.56 for M1. Top-left: diversity entropy is comparable across mechanisms. Bottom-right: the gate trades a modest rise in exposure Gini for the large quality gain—an effect consistent with the pipeline-depth concentration seen in H4.

5.1 H2 (headline): a quality gate moves the equilibrium to quality, across all seeds

Comparing M3 against the M1 baseline, the quality gate produces a large and reproducible shift in the creator equilibrium. Across all five seeds, M3's final mean bait settles in the range ∼ 0.03–0.09, versus ∼ 0.46–0.67 under M1; and M3's final mean quality settles in the range ∼ 0.87–0.97, versus ∼ 0.40–0.56 under M1. In words: once the pipeline screens content for quality before reranking on engagement, creators stop investing in bait and start investing in quality, because bait no longer survives the gate to earn impressions. This is the strongest, most robust result in the paper and the one I would stake the project's credibility on. It is also the most actionable: it says that a relatively cheap, interpretable screening step placed *upstream* of the engagement objective can change what creators produce, not merely what users see. (Confirmed 5/5 seeds.) One honest caveat is visible in the exposure panel of Figure 1: the gate does not come free on the fairness axis—M3's exposure Gini sits somewhat *above* M1's, because concentrating impressions on the creators who clear the quality bar is itself a form of concentration. The gate buys a large, robust quality gain at the cost of a modest rise in exposure inequality, rather than improving both at once.

5.2 H1: robust Gini collapse, seed-fragile bait inflation

The pre-registered hypothesis was that IPPO creators under a pure engagement ranker (M1) converge to low-diversity, high-bait, winner-take-all content. The data force a more careful

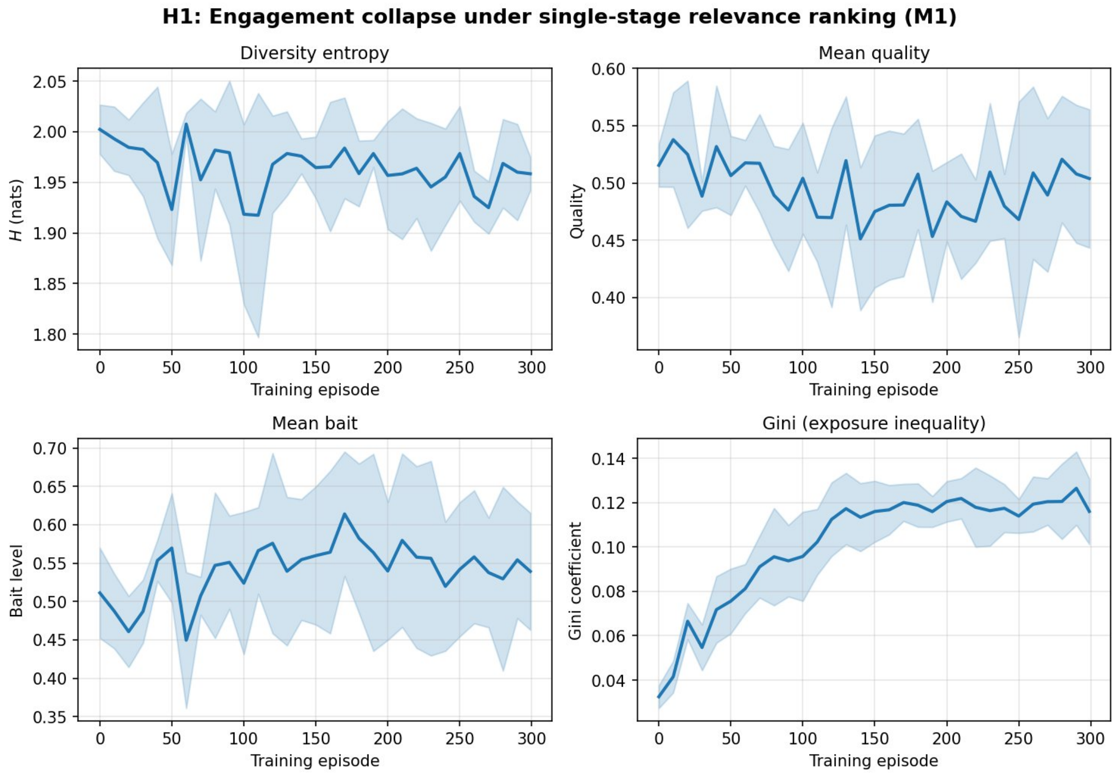

Figure 2: H1 (5 seeds, shaded = across-seed spread). Under pure engagement ranking (M1). Bottom-right: exposure Gini rises robustly on every seed (mean +263%, s.d. 59), climbing from ~ 0.03 to a ~ 0.12 plateau. Bottom-left: mean bait rises on the aggregate but the effect is seed-fragile—driven by seed 42 (+44%) while seeds 7, 13, and 99 fall (see text); the wide band reflects exactly this disagreement. Top row: diversity entropy and mean quality stay roughly flat/noisy, so the headline story is concentration of exposure, not loss of nominal diversity.

statement, and the split is the methodological heart of the paper.

The exposure-concentration effect is robust. Across all five seeds, the Gini coefficient of creator exposure rises sharply over training—every seed passes a +50% threshold, and the mean increase is about +263% (standard deviation 59). A rich-get-richer dynamic in *who gets seen* is a reliable consequence of engagement ranking in this market.

The bait-inflation effect is not robust. Only seed 42 shows the textbook rise in bait (about +44%, clearing a +30% threshold); seeds 7, 13, and 99 actually show bait *decreasing* over training. I therefore explicitly decline to report "engagement ranking causes a clickbait collapse" as a general result. What generalises is the concentration of exposure; whether the concentrated winners happen to be bait-heavy is seed-dependent in this setup. Presenting these two effects as a single tidy collapse would misrepresent the evidence, and the distinction is exactly the sort that re-seeding exists to expose.

5.3 H4 (single seed, exploratory): pipeline depth concentrates exposure

On seed 42 only, the two-stage mechanism M2 yields a final exposure Gini of 0.187 versus 0.092 for single-stage M1—roughly a +203% increase in inequality. This is consistent with the component-interaction story of Hron et al. (2021): adding a retrieval stage and an online reranker is not a neutral refinement; the stages interact to concentrate exposure on a shrinking set of creators. Because this is a single seed, I report it as a directional, exploratory finding with no error bars and make no significance claim.

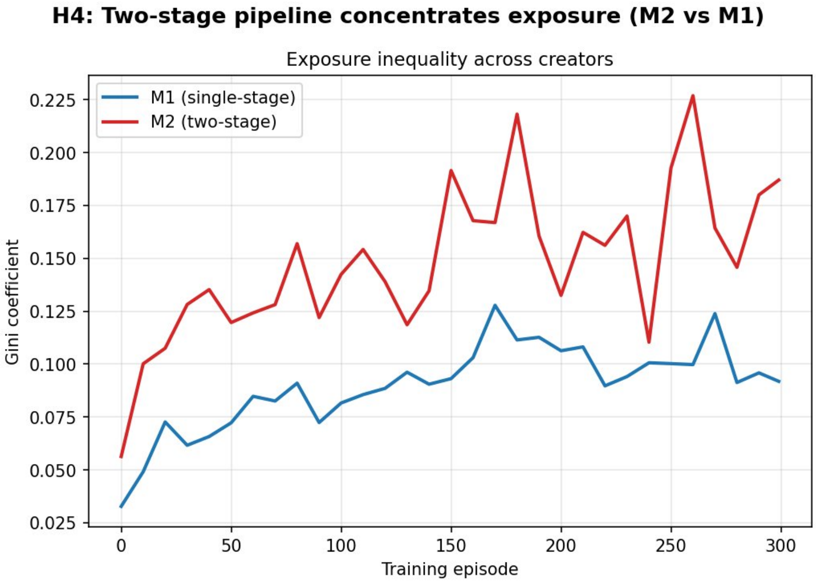

Figure 3: H4 (exploratory, seed 42). The two-stage pipeline (M2) nearly triples final exposure Gini relative to single-stage (M1): 0.187 vs 0.092.

5.4 H3 (single seed, exploratory): recommender temperature governs specialisation, visible through bait

I swept the recommender temperature over $\{0.1, 0.3, 1, 3, 10\}$ on seed 42. The pre-registered prediction was that temperature would govern specialisation, readable in `diversity_entropy`. That prediction is only weakly borne out in entropy itself: diversity entropy stays in a narrow ~ 1.84–2.0 band and moves non-monotonically across temperatures, so I do not lean on it. The signal is instead clear and strong in bait. Final mean bait rises from about 0.12 at $\tau = 0.1$, to about 0.82 at $\tau = 1$, to roughly 0.99 at $\tau \in \{3, 10\}$. The interpretation: a sharp ranker (low temperature) makes the relevance inner-product term dominate the click logit in Eq. (1), so creators are forced to match user preferences precisely—specialisation by relevance, low bait. A flat ranker (high temperature) flattens relevance differences, so the bait term dominates the logit and creators pile into bait because nothing else moves the needle. Temperature, in other words, is a lever on how hard the mechanism pushes creators toward genuine relevance-matching versus cheap engagement. Again: single seed, exploratory, no error bars.

6 Discussion

The four studies tell a coherent mechanism-design story even though they differ in robustness. Left to a pure engagement objective (M1), the market reliably concentrates exposure (H1's robust Gini collapse) and, depending on the seed, may also drift toward bait (H1's fragile bait effect; H3 shows that a flat ranker makes the bait drift near-certain). Adding architectural depth without changing the objective (M2) makes concentration worse, not better (H4). The intervention that actually changes creator behaviour for the better is a *quality gate placed upstream of the engagement objective* (M3, H2): it is the only manipulation here that robustly pulls the equilibrium toward quality and away from bait, across every seed.

The practical reading is that *where* you put a quality signal matters as much as whether you have one. An engagement reranker, however sophisticated, optimises the proxy it is given; a gate that filters on quality before the reranker ever sees the candidates changes the creators' best response, because bait that cannot pass the gate cannot earn impressions and therefore

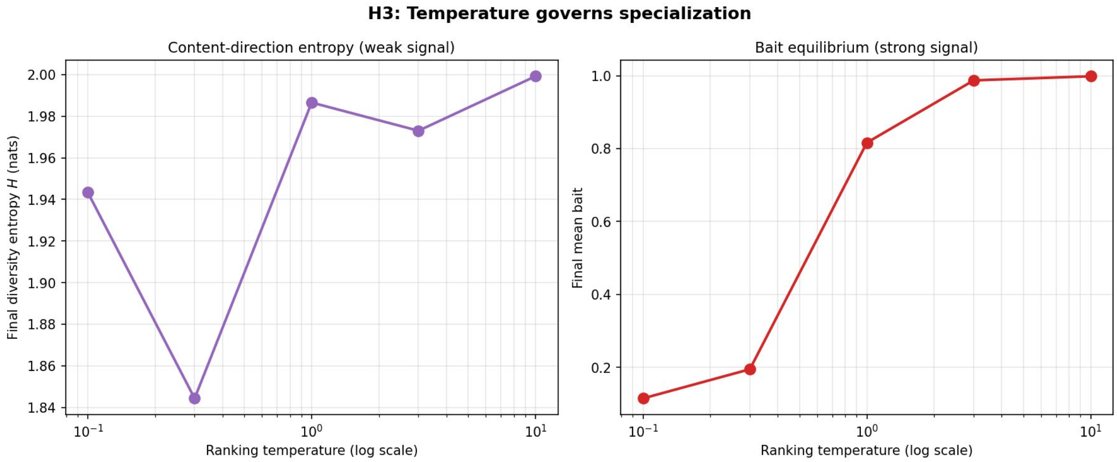

Figure 4: H3 (exploratory, seed 42). Final-equilibrium values versus ranking temperature (log scale). Left: diversity entropy is a weak, non-monotonic signal, staying within ~ 1.84–2.0. Right: final mean bait is a strong, near-monotone signal, rising from ~ 0.12 at $\tau = 0.1$ through ~ 0.82 at $\tau = 1$ to ≈ 0.99 at $\tau \in \{3, 10\}$—a sharp ranker forces relevance-matching (low bait), a flat ranker lets the bait term dominate.

cannot pay. This is consistent with the theoretical intuition from the strategic-creator literature that the equilibrium content distribution is a function of the mechanism, not just of user preferences—and it gives an operational handle on that intuition. It also dovetails with the cautionary result of Immorlica et al. (2024) that, under engagement optimisation, quality and gaming are positively correlated and average consumed quality can fall: my M3 result suggests that *moving the quality check off the engagement path* is one way to break that coupling.

It is worth dwelling on the honesty point in H1, because it is the part most likely to be quietly "cleaned up" in a less careful write-up. The seductive version of this paper claims that engagement ranking causes a clickbait collapse and shows seed 42 as proof. The data do not support that as a general claim. Three of five seeds move bait the other way. What survives re-seeding is the exposure concentration. Keeping those two effects separate—and labelling the bait effect as fragile—is the difference between a result and a just-so story.

7 Limitations

Three limitations are load-bearing and I state them plainly.

First, **M3 is not a live LLM.** The quality gate is `SimpleLinearGate` with $\alpha_{\text{quality}} = 3$, $\alpha_{\text{bait}} = -3$—an analytical linear proxy for a quality judge. A live `LLMQualityGate` and a `DistilledGate` exist in the codebase but were not run. Every M3 number in this paper is from the linear gate, and the LLM framing is a conceptual stand-in, not an empirical claim about language-model judges. Real LLM judges bring their own well-documented biases (position, verbosity, self-enhancement; Zheng et al., 2023; Gu et al., 2024), which a linear gate does not model.

Second, **only H1 and H2 carry multi-seed evidence.** Those two studies use five seeds and I report spread (e.g. H1's Gini mean +263%, s.d. 59). H4 and H3 are single-seed (seed 42). They are exploratory and directional; I attach no error bars and make no significance claims for them.

Third, **the scale is small and the retrieval is exact.** Twenty creators and fifty users is a deliberately small market, chosen for controllability and fast iteration, not for external validity to platform scale. Retrieval uses NumPy inner-product search rather than FAISS—an intentional choice to dodge a PyTorch+FAISS OpenMP crash on macOS, and mathematically equivalent to exact nearest-neighbour search at this scale, but not a test of approximate-retrieval effects.

8 Future Work

The obvious next step is to make M3 real: run the live `LLMQualityGate` and the `DistilledGate` path, and check whether a learned judge (and a distilled, cheap approximation of it) reproduces the linear gate's equilibrium shift—and what new failure modes its biases introduce. Second, the exploratory studies deserve multi-seed confidence intervals: re-running H3 and H4 across the full seed set $\{42, 0, 7, 13, 99\}$ would tell me whether the temperature→bait curve and the pipeline-depth→Gini effect are as robust as H2 or as fragile as H1's bait. Third, scaling up the market—hundreds of creators, thousands of users, a real ANN index—would test whether the qualitative mechanism effects survive at a scale closer to deployment, and would let me revisit the specialisation and genre-formation predictions of Jagadeesan et al. (2023) that the small market only weakly exercises.

9 Conclusion

Treating the recommender as a mechanism and creators as learning agents turns "what should we rank?" into "what content are we incentivising into existence?" In a small but fully controlled market, the answer depends sharply on the mechanism. Pure engagement ranking robustly concentrates exposure and, depending on conditions, can tip creators toward bait. A quality gate placed upstream of the engagement objective robustly moves the equilibrium toward quality—the clearest, most reproducible effect I found. The contribution is a reproducible testbed plus a disciplined account of which effects survive re-seeding and which do not, including an explicit refusal to oversell a fragile bait result. I hope the honest version is more useful than the tidy one.

Code, configs, results logs, and figures: `https://github.com/ozeidan9/recsim-arena`.

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